



Middle Cerebral Artery Aneurysm Trial (MCAAT): A Randomized Care Trial Comparing Surgical and Endovascular Management of MCA Aneurysm Patients

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■ **BACKGROUND:** Whether the best management of middle cerebral artery (MCA) aneurysm patients is surgical or endovascular remains uncertain, with little evidence to guide decision-making. A randomized care trial offering MCA aneurysm patients a 50% chance of surgical and a 50% chance of endovascular management may optimize outcomes in the presence of uncertainty.

■ **METHODS:** The Middle Cerebral Artery Aneurysm Trial (MCAAT) is an investigator-initiated, multicenter, parallel group, prospective, 1:1 randomized controlled clinical trial. All adult patients with MCA aneurysms, ruptured or unruptured, amenable to surgical and endovascular treatment can be included. The composite primary outcome is "Treatment Success": (i) occlusion or exclusion of the aneurysm using the allocated treatment modality; (ii) no intracranial hemorrhage during follow-up; (iii) no retreatment of the target aneurysm during follow-up; (iv) no residual aneurysm on angiographic follow-up; and (v) independence (mRS <3) at 1 year. The trial tests 2 versions of the same hypothesis (one for ruptured and one for unruptured MCA aneurysm patients): Surgical management

will lead to a 15% absolute increase in the proportion of patients reaching Treatment Success from 55% to 70% (ruptured) or from 75% to 90% (unruptured aneurysm patients) compared with endovascular treatment (any method). In this pragmatic trial, outcome evaluations are by treating physicians, except for 1-year angiographic results which will be core lab assessed. The trial will be monitored by an independent data safety monitoring committee to assure safety of participants. MCAAT is registered at [clinicaltrials.gov: NCT05161377](https://clinicaltrials.gov/ct2/show/study/NCT05161377).

■ **CONCLUSIONS:** Patients with MCA aneurysms can be optimally managed within a care trial protocol.

INTRODUCTION

Intracranial aneurysms are increasingly being treated with endovascular methods. However, many middle cerebral artery (MCA) aneurysm patients are still managed surgically, and many surgeons consider aneurysms in this location to be a distinct

Key words

- Endovascular management
- MCA aneurysm
- Randomized trial
- Surgical clipping

Abbreviations and Acronyms

BRAT: Barrow Ruptured Aneurysm Trial
CI: Confidence interval
CURES: Canadian UnRuptured Endovascular Versus Surgery
ISAT: International Subarachnoid Aneurysm Trial
ISFD: Intra-saccular flow disruptor
MCA: Middle cerebral artery
MCAAT: Middle Cerebral Artery Aneurysm Trial
mRS: modified Rankin Scale score
RA: Ruptured aneurysm
RCT: Randomized controlled trial
UIA: Unruptured intracranial aneurysm
WEB: Woven EndoBridge

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subgroup for which clipping may still be best.^{1,2} However, convincing randomized evidence supporting the use of microsurgical clipping is lacking.

Anatomic features of MCA aneurysms, such as a wide neck often including a branch artery origin, render endovascular treatment difficult.^{3,4} New devices have been and continue to be developed to address these difficulties, including stents,⁵ flow diverters,⁶ and intra-saccular flow disruptors (ISFDs),⁷ but none have been shown to improve endovascular results in a randomized trial.⁸⁻¹⁰

The International Subarachnoid Aneurysm Trial (ISAT) has shown that small anterior circulation good-grade ruptured aneurysm patients have better 1-year clinical outcomes after being coiled than clipped,¹¹ but those results may not necessarily apply to patients with MCA aneurysms. Between 1994 and 2001, the ISAT recruited a small number of selected MCA aneurysm patients—only those eligible for the sole endovascular option available at the time, simple coiling. Poor clinical outcomes (modified Rankin Scale score [mRS] >2, the primary endpoint at 1 year) were similar between groups, occurring in 39 of 139 clipped (28.1%, 95% confidence interval [CI]: 21%–36%), and 46 of 162 coiled patients (28.4%, 95% CI: 22%–36%).¹² The suspicion that ISAT MCA aneurysm patients were selected is supported by the pre-randomized Barrow Ruptured Aneurysm Trial (BRAT) study: Of 61 MCA aneurysm patients included in BRAT between 2003 and 2007, 21 of the 31 (68%) patients assigned to coiling were crossed-over to clipping.¹³ The Finnish randomized controlled trial (RCT) comparing clipping and coiling similarly reported only 19 ruptured MCA aneurysms because 59 other patients were excluded.^{14,15}

For unruptured MCA aneurysms, comparative case series suggest that clipping may lead to better angiographic outcomes¹⁶ but all available data are subject to the shortcomings of observational studies. One exception is the Canadian UnRuptured Endovascular Versus Surgery (CURES) randomized trial¹⁷: Exploratory analyses showed that better efficacy may be obtained with surgical than endovascular management of unruptured MCA aneurysms, but subgroup results should always be interpreted with caution.^{1,18}

With patients and clinicians drawn towards novel and less invasive endovascular options, surgical clipping has been described as “fighting an uphill battle with fewer and fewer troops”.¹⁹ Expert opinions and case series will not be sufficient to justify open neurosurgical aneurysm repair in the future. There are ongoing comparative trials addressing the best treatment of unruptured intracranial aneurysms (UIAs) and ruptured aneurysms (RAs) in all locations,^{17,20} but these all-inclusive trials do not take into consideration the specific nature of MCA aneurysms. For example, ISAT-2, a continuation of ISAT, hypothesizes that endovascular treatment is superior to clipping.²⁰ If clipping of MCA aneurysms turns out to be superior, interpretation of subgroup results contrary to the overall trial hypothesis would be problematic.²¹ Randomized trials testing the hypothesized superiority of clipping in this particular and frequent location are needed for both ruptured and unruptured MCA aneurysm patients.¹

The goals of the Middle Cerebral Artery Aneurysm Trial (MCAAT) are (1) to compare results of surgical and endovascular management of MCA aneurysms, ruptured and unruptured and (2) while waiting for randomized evidence, to provide a care trial context to optimize clinical outcomes in the meantime.²²

Trial Design

MCAAT provides a transparent care trial context for clinicians, despite the serious uncertainty, to optimally manage patients with MCA aneurysms, offering each patient a 50% chance of being treated with surgical clipping and a 50% chance of being treated with more recent, less-invasive but unproven endovascular options.

MCAAT is a parallel group 1:1 pragmatic care trial comparing clipping of MCA aneurysms to any endovascular management option (including coiling, stent-assisted coiling, flow diversion, ISFDs (such as the Woven EndoBridge device [WEB], MicroVention, Aliso Viejo, California, USA), or other novel device). The lack of convincing evidence in the neurovascular literature does not allow for rigid treatment protocols; the choice of the endovascular comparator intervention will be according to local expertise and clinical judgment, but prespecified prior to randomized allocation.

MCAAT addresses 2 versions of the same clinical dilemma, one concerning ruptured, the other unruptured, MCA aneurysm patients, but to reduce bureaucracy and simplify organization and to minimize delays and costs, both will be studied within one trial. The MCAAT trial is registered at [clinicaltrials.gov: NCT05161377](https://clinicaltrials.gov/ct2/show/study/NCT05161377).

Selection Criteria

MCAAT is an all-inclusive pragmatic care trial: Selection criteria are inclusive and exclusions few: Patients are at least 18 years of age; they harbor at least 1 MCA aneurysm, ruptured or unruptured (an aneurysm that is suspected to have ruptured more than 30 days prior to study inclusion will be considered unruptured). Aneurysm remnants or recurrences after treatment can also be included if deemed appropriate and eligible for both options; the patient and aneurysms are considered appropriate for either surgical or endovascular treatment (using any method or approved device) by the treating team. Excluded patients are those with absolute contraindications to the administration of contrast material, patients with arteriovenous malformation—associated aneurysms, poor grade (World Federation of Neurological Surgeons Grading Scale 5) ruptured aneurysm patients, and when informed consent cannot be obtained.

Management Options

Surgical clipping or endovascular treatment are performed according to local standards of practice. Details regarding surgical or endovascular management such as the use of coils, balloon-remodeling, stents, flow-diverters, ISFDs or other innovative devices, antiplatelet regimens and coagulation or platelet aggregation testing, as well as post-treatment medical management issues, are recorded.

Practical Arrangements for Allocating Participants to Treatment Groups

MCAAT centers are expected to routinely use surgical clipping or endovascular treatment for MCA aneurysms. Potential trial participants will be identified at the time of a multidisciplinary meeting (unruptured aneurysm patients), but patients with ruptured MCA aneurysms can be recruited at any time. All MCA aneurysm patients will be registered but randomized allocation of treatments will be performed when eligibility for both treatment options can be determined (and can be performed urgently for ruptured aneurysm patients) by local endovascular and surgical

experts. After online registration, only strictly necessary data will be recorded, and participating clinicians will pre-specify the comparator endovascular treatment that would be used in case of endovascular allocation (coiling with or without balloon remodeling, with or without stenting, flow diversion, WEB/ISFD, or other). Patients judged to be untreatable by any endovascular option (expected to be few given the all-inclusive approach) will be entered in a parallel registry. Patients are then randomly allocated to be treated with: (a) surgical management or (b) endovascular management. Wherever approved by institutional research committees, the trial will use pre-randomization.²³ All patients or legal substitutes will be informed of the randomized nature of the trial at the time of informed consent. In centers where pre-randomization has been approved, they will be encouraged to participate in the RCT but “cross-overs” will be recorded. The decision to not participate in the research study at all will be respected, and the number of patients excluded will be recorded in an anonymous case-log. Stratified randomized allocation is performed separately for ruptured and for unruptured aneurysm patients, using a Web-based platform available 24 hours/day to ensure concealed allocation. Minimization is used to balance 2 risk factors: (i) age >65 years at time of inclusion, and (ii) aneurysm size < or ≥10 mm (using maximum aneurysm diameter on cross-sectional imaging).

Primary Outcome Measure

The primary outcome measure is Treatment Success, defined as a composite safety and efficacy endpoint consisting of: (i) occlusion or exclusion of the aneurysm using the allocated treatment modality; (ii) no intracranial hemorrhage during follow-up, documented by computed tomography scanning or positive lumbar puncture in the setting of acute headache; (iii) no retreatment of the target aneurysm during follow-up, (iv) no residual aneurysm at 1 year (defined according to a published classification system with proven reliability²⁴ and adjudicated by an independent core lab); and (v) independent functional status (mRS <3) at 1 year using a standardized questionnaire.²⁵ Patients “not meeting” more than one of the items of the composite endpoint are counted once, and will be reported in the following hierarchical order: Failure to exclude the aneurysm using the allocated treatment modality; intracranial hemorrhage during follow-up; re-treatment during follow-up; mRS >2 at 1 year; residual aneurysm at 1 year.

Secondary Endpoints

Secondary endpoints will include the individual elements of the composite primary endpoint, in addition to: peri-treatment hospitalization lasting more than 5 days (UIAs) or 20 days (RAs); peri-operative morbidity and mortality (mRS >2 at 1 month); discharge following treatment to a location other than home; overall morbidity and mortality at 1 and 5 years (all causes); aneurysm re-rupture following randomization but before treatment initiation.

Frequency and Duration of Follow-up

All of the investigations proposed in MCAAT constitute part of routine, standard care in the treatment of unruptured or ruptured intracranial aneurysms. Patients from both trial arms will be seen in clinic at approximately 6–8 weeks post-treatment as part of routine follow-up care. Patients will be followed with routine clinic

visits at 1 year and every 2 years thereafter. At these intervals, contact information will be verified, mRS scores will be determined according to a standardized questionnaire, and inquiries will be made about aneurysm rupture, re-treatment, or hospital admission for any reason. All patients will have catheter or non-invasive angiography according to local practice (computed tomography angiography or magnetic resonance angiography) at 12 ± 2 months and at 5 years ± 1 year post-treatment to determine the presence of a major aneurysm recurrence (Figure 1).

Trial Hypotheses and Number of Patients to Be Recruited

There are 2 separate versions of the same primary trial hypothesis, one for unruptured and one for ruptured aneurysms (mainly because the mRS status of ruptured MCA aneurysm patients is expected to differ from unruptured MCA aneurysm patients). For unruptured aneurysms, the primary hypothesis is: Surgical management will lead to an increase in the proportion of patients reaching the primary endpoint of treatment success from 75% to 90% compared with endovascular treatment (any endovascular method). With target alpha 0.05, and power 80%, 280 UIA patients (139 per group, no losses) would be sufficient to demonstrate a significant difference (2-sided Fisher's exact test). We expect 10% of patients to be lost to follow-up. The target number of UIA patients to be recruited would therefore be approximately 310. These estimates (and the 15% difference) were inspired from published results of a randomized comparison of MCA aneurysm patients extracted from the CURES study, reporting a 20% difference in angiographic outcomes between endovascular and surgical treatment of unruptured MCA aneurysms at one year.^{1,20}

For ruptured aneurysms, the primary hypothesis is: Surgical management will lead to an increase in the proportion of patients reaching the primary endpoint of treatment success from 55% to 70% compared with endovascular treatment. With target alpha 0.05, and power 80%, 350 patients (175 per group, no losses) would be sufficient to demonstrate a significant difference (2-sided Fisher's exact test). With some losses to follow-up expected (10%), the target number of patients to be recruited would therefore be approximately 385. These estimates (and the 15% difference) were inspired from published results of a randomized comparison of MCA aneurysm patients extracted from the ISAT-2 study, reporting a 15% difference in clinical (mRS > 2) and angiographic outcomes between endovascular and surgical treatment of ruptured MCA aneurysms at 1 year.¹

We cannot precisely anticipate the proportion of ruptured and unruptured aneurysm patients that will be recruited, but a 50%/50% estimate leads to a total number of patients to be recruited on the order of 700.

We anticipate that for centers where pre-randomization will be used a certain number of patients (or surrogates) will disagree with the allocated management option and “cross-over” to the alternative option. Because analyses will be intent to treat, as many as 1000 patients may be required to reach significant conclusions for both ruptured and unruptured subgroups.

Participating Centers

Any center routinely using surgical clipping or endovascular treatment (any method) for ruptured and unruptured MCA aneurysms can participate in MCAAT.

Evaluation	Screening	Pre-entry	Entry	Treatment	Discharge	6 months	1 year	5 years
Documentation of MCA aneurysm	X							
Imaging*	X						X	
Informed consent			X					
Medical history		X						
Clinical assessment		X			X		X	
Neurologic exam		X			X		X	X
mRS		X			X		X	X
Failure to occlude aneurysm				X				
Number of days in hospital					X			
Discharge disposition					X			
Residual aneurysm on FU imaging							X	X
Hemorrhage during FU						X	X	X
Telephone interview								X
* Including catheter angiography or non-invasive imaging.								

Figure 1. Schedule of evaluations. FU, follow-up.

Statistical Analyses

Descriptive statistics will be done on demographic variables and preoperative data to compare the 2 groups at baseline. Means, standard deviations, and range will be presented for quantitative variables and frequency tables for categorical variables. Those statistics will be broken down by presentation (ruptured or unruptured) and by treatment arm. The primary outcome will be compared between surgical or endovascular management using regression analysis at 1 year in intent-to-treat analyses. Predefined subgroup analyses will include stratified presentation criteria (RA or UIA) as well as stratified randomized allocation according to the choice of the pre-specified endovascular strategy, in addition to minimization criteria (aneurysm size and age). The same analyses will be also performed for “as treated” groups.

Secondary outcomes will be compared between groups using independent *t* tests (quantitative variables) and χ^2 tests (categorical variables). All tests will be interpreted with a 0.05 level of confidence.

Regulatory and Ethical Considerations

Participants will be made fully aware of the study purposes and procedures. When signing the study consent form, they will be informed that participation is voluntary, and that they may withdraw from the trial at any point. Patient enrollment in this trial will comply with the principles enunciated in the Declaration of Helsinki. All collected information will be anonymized and kept confidential. Trial management will be transparent, independent, and will aim to preserve the scientific integrity of the research and the well-being of the participants. The Steering Committee will not have access to the unmasked data before completion or

interruption of the trial. The Data Safety and Monitoring Committee will follow the progress of the trial, with masked results and events, but with the possibility of unmasking if necessary.

DISCUSSION

The less invasive endovascular approach has revolutionized the management of intracranial aneurysms, but the crucial question is whether it actually translates into better patient outcomes. There is evidence from ISAT that coiling improves outcomes for many ruptured aneurysms, but this may not be the case for MCA aneurysms.^{1,11,17}

The fundamental question of whether or not unruptured aneurysms should be treated at all will not be addressed by this trial.^{26,27} An interim examination of the unruptured MCA aneurysm patients included in the CURES trial showed that coiling led to a statistically significantly higher incidence of treatment failure (13 of 48 [27%; 95% CI: 0.17–0.41] compared with 3 of 42 [7%; 95% CI: 0.02–0.19] for clipping, *P* = 0.025).¹ However, the primary outcome from the CURES trial was a composite that primarily included a surrogate angiographic outcome measure.¹⁷ All other published comparisons of endovascular and open surgical techniques for unruptured MCA aneurysms are non-randomized and therefore subject to all of the biases of observational studies.

Many MCA aneurysms present endovascular challenges for which innovative devices have been introduced over the last decade, however many neurovascular centers still regard microsurgical clipping as the treatment of choice.^{2,28–30} The current problem is 2-fold: How can a surgical treatment, known to be more invasive and to cause more immediate (even if transient) neurological deficits, be proposed when convincing evidence of its

relative merit is lacking? Yet, endovascular innovations have been adopted on the basis of enthusiastic single-arm case series, and no one knows if their use is in the patients' best interests. In this context, MCAAT proposes that optimal care is to offer each patient a 50% chance of surgical clipping combined with a 50% chance of endovascular treatment, within a care trial designed in the best interest of MCA aneurysm patients.

The price to pay for the design choice of including multiple endovascular comparators is that MCAAT will not be able to determine which of the various endovascular treatment options is responsible for the result, good or bad, that will be observed in the endovascular group. This pragmatic choice was considered preferable to the design of multiple explanatory trials comparing clipping to each endovascular option, which would have necessitated numerous specific selection criteria, in the absence of evidence and on which neurovascular specialists disagree.^{31,32} Furthermore, this "explanatory" approach³³ would still ultimately leave the community without a reliable means to compare results between trials. It will be possible to analyze subgroup results per-particular endovascular technique, defined according to intention to treat or to treatment actually received, but a valid comparison between endovascular techniques will not be possible. The final results of MCAAT could either show that surgical management of MCA aneurysms is superior to any endovascular option, or that it is not. There can be no clear single endovascular technique declared winner or loser in this proposed trial.

Publication of research protocols informs the community of clinicians and patients that the study is ongoing, the importance of which has been emphasized by journal editors, peer reviewers, researchers, and public advocates.³⁴ Publication may also promote recruitment of interested centers and participants. It also allows the trial design to be openly discussed and criticized by the medical or surgical community that is concerned with the clinical dilemma. Finally, the publication of a research protocol is a healthy process no matter the end result, because it transparently exposes the uncertainty regarding the respective merits of the interventions being tested in the study.

Surgical RCTs are difficult to design, organize, implement and complete.³⁵ Beyond the search for a final answer, we must attempt to provide our patients with the best possible care in spite of the uncertainty regarding whether surgical or endovascular management is best. MCAAT has been designed as a care trial integrated into routine practice.^{36,37} Care trials are real-world studies that test the relative merits and drawbacks of treatments used in everyday medical and surgical practices, while at the same time optimizing outcomes for participating patients. The end-points chosen for care trials should be simple, clinical, meaningful, and resistant to bias. Follow-up visits and tests are part of

routine care. Data are collected on simple electronic forms. The primary goal of a care trial is to provide prudent treatments in a context where evidence is lacking.³⁶ The simplicity of the study ensure that the trial can be completed while maintaining costs and delays within reasonable limits.

We elected to include a surrogate angiographic outcome within the primary outcome measure because clinical care remains driven by the intent to protect patients from future aneurysm ruptures, events that are anticipated by angiographic results and without which decades of clinical follow-up would be required. Thus the primary clinical and angiographic composite outcome "Treatment Success" is relevant to patients and clinicians. It also allows for a trial hypothesis that can be confirmed or refuted with a reasonable number of patients followed for a reasonable time period. The category of "residual aneurysm" was chosen for the angiographic verdict, for it was shown to be more reliably made between observers, and had a more constant clinical meaning than other categories, but "complete occlusions" will also be examined in secondary analyses.^{24,38}

CONCLUSIONS

Whether ruptured or unruptured MCA aneurysm patients are best treated surgically or by endovascular methods remain hypotheses that must be tested. The MCAAT trial provides a care trial context for clinicians to best manage their patients in the face of uncertainty.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Tim E. Darsaut: Conceptualization, Data curation, Investigation, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing. **Michael B. Keough:** Data curation, Investigation. **William Boisseau:** Data curation, Investigation, Methodology. **J. Max Findlay:** Data curation, Investigation. **Michel W. Bojanowski:** Data curation, Investigation. **Chiraz Chaalala:** Data curation, Investigation. **Daniela Iancu:** Data curation, Investigation. **Alain Weill:** Data curation, Investigation. **Daniel Roy:** Data curation, Investigation. **Laurent Estrade:** Data curation, Investigation. **Jean-Paul Lejeune:** Data curation, Investigation. **Anne-Christine Januel:** Data curation, Investigation. **Andrew P. Carlson:** Data curation, Investigation. **Eric Sauvageau:** Data curation, Investigation. **Hosam Al-Jehani:** Data curation, Investigation. **Kirill Orlov:** Data curation, Investigation. **Sorin Aldea:** Data curation, Investigation. **Michel Piotin:** Data curation, Investigation. **Thomas Gaberel:** Data curation, Investigation. **Guylaine Gevry:** Data curation, Investigation, Project administration, Resources, Supervision. **Jean Raymond:** Conceptualization, Data curation, Investigation, Methodology, Resources, Supervision, Writing – original draft, Writing – review & editing.

REFERENCES

- Darsaut TE, Keough MB, Sagga A, et al. Surgical or endovascular management of middle cerebral artery aneurysms: a randomized comparison. *World Neurosurg.* 2021;149:e521-e534.
- Berro DH, L'Allinec V, Pasco-Papon A, et al. Clip-first policy versus coil-first policy for the exclusion of middle cerebral artery aneurysms. *J Neurosurg.* 2019;20:1-8.
- Iijima A, Piotin M, Mounayer C, Spelle L, Weill A, Moret J. Endovascular treatment with coils of 149 middle cerebral artery berry aneurysms. *Radiology.* 2005;237:611-619.
- Vanzin JR, Mounayer C, Piotin M, Spelle L, Boissonnet H, Moret J. [Endovascular treatment of unruptured middle cerebral artery aneurysms]. *J Neuroradiol.* 2005;32:97-108.
- Zhang X, Zuo Q, Tang H, et al. Stent assisted coiling versus non-stent assisted coiling for the management of ruptured intracranial aneurysms: a meta-analysis and systematic review. *J Neurointerv Surg.* 2019;11:489-496.
- Cagnazzo F, Mantilla D, Lefevre PH, Dargazanli C, Gascou G, Costalat V. Treatment of middle

- cerebral artery aneurysms with flow-diverter stents: a systematic review and meta-analysis. *AJNR Am J Neuroradiol*. 2017;38:2289-2294.
7. Pierot L, Szikora I, Barreau X, et al. Aneurysm treatment with WEB in the cumulative population of two prospective, multicenter series: 3-year follow-up. *J Neurointerv Surg*. 2021;13:363-368.
 8. Raymond J, Darsaut TE, Guilbert F, Weill A, Roy D. Flow diversion in aneurysms trial: the design of the FIAT study. *Interv Neuroradiol*. 2011;17:147-153.
 9. Fahed R, Darsaut TE, Raymond J. The introduction of innovations in neurovascular care: patient selection and randomized allocation. *World Neurosurg*. 2018;118:e99-e104.
 10. Chapot R, Mosimann PJ, Darsaut TE, Raymond J. Retirements must be included in the evaluation of device performance. *J Neurointerv Surg*. 2021;13:e5.
 11. Molyneux A, Kerr R, Stratton I, et al. International Subarachnoid Aneurysm Trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised trial. *Lancet*. 2002;360:1267-1274.
 12. Molyneux AJ, Kerr RS, Yu LM, et al. International Subarachnoid Aneurysm Trial (ISAT) of neurosurgical clipping versus endovascular coiling in 2143 patients with ruptured intracranial aneurysms: a randomised comparison of effects on survival, dependency, seizures, rebleeding, subgroups, and aneurysm occlusion. *Lancet*. 2005;366:809-817.
 13. Mooney MA, Simon ED, Brigeman S, et al. Long-term results of middle cerebral artery aneurysm clipping in the Barrow Ruptured Aneurysm Trial. *J Neurosurg*. 2018;130:895-901.
 14. Koivisto T, Vanninen R, Hurskainen H, Saari T, Hernesniemi J, Vapalahti M. Outcomes of early endovascular versus surgical treatment of ruptured cerebral aneurysms. A prospective randomized study. *Stroke*. 2000;31:2369-2377.
 15. Vanninen R, Koivisto T, Saari T, Hernesniemi J, Vapalahti M. Ruptured intracranial aneurysms: acute endovascular treatment with electrolytically detachable coils—a prospective randomized study. *Radiology*. 1999;211:325-336.
 16. Smith TR, Cote DJ, Dasenbrock HH, et al. Comparison of the efficacy and safety of endovascular coiling versus microsurgical clipping for unruptured middle cerebral artery aneurysms: a systematic review and meta-analysis. *World Neurosurg*. 2015;84:942-953.
 17. Darsaut TE, Findlay JM, Magro E, et al. Surgical clipping or endovascular coiling for unruptured intracranial aneurysms: a pragmatic randomised trial. *J Neurol Neurosurg Psychiatry*. 2017;88:663-668.
 18. Rothwell PM. Treating individuals 2. Subgroup analysis in randomised controlled trials: importance, indications, and interpretation. *Lancet*. 2005;365:176-186.
 19. Macdonald RL. Editorial: Clip or coil? Six years of follow-up in BRAT. *J Neurosurg*. 2015;123:605-607.
 20. Darsaut TE, Roy D, Weill A, et al. A randomized trial of endovascular versus surgical management of ruptured intracranial aneurysms: Interim results from ISAT2. *Neuro-Chirurgie*. 2019;65:370-376.
 21. Magro E, Gentric JC, Darsaut TE, Ziegler D, Bojanowski MW, Raymond J. Responses to ARUBA: a systematic review and analysis for the design of future AVM trials. *J Neurosurg*. 2017;126:486-494.
 22. Darsaut TE, Raymond J. Ethical care requires pragmatic care research to guide medical practice under uncertainty. *Trials*. 2021;22:143.
 23. Raymond J, Darsaut TE, Roy DJ. Recruitment in clinical trials: the use of Zelen's prerandomization in recent neurovascular studies. *World Neurosurg*. 2017;98:403-410.
 24. Benomar A, Farzin B, Volders D, et al. Angiographic results of surgical or endovascular treatment of intracranial aneurysms: a systematic review and inter-observer reliability study. *Neuroradiology*. 2021;63:1511-1519.
 25. Bruno A, Shah N, Lin C, et al. Improving modified Rankin Scale assessment with a simplified questionnaire. *Stroke*. 2010;41:1048-1050.
 26. Darsaut TE, Desal H, Cognard C, et al. Comprehensive Aneurysm Management (CAM): an all-inclusive care trial for unruptured intracranial aneurysms. *World Neurosurg*. 2020;141:e770-e777.
 27. Darsaut TE, Fahed R, Raymond J. Unruptured aneurysms: why observational studies fall short no matter how "big" the data. *Neurochirurgie*. 2021;67:330-335.
 28. Nussbaum ES, Madison MT, Goddard JK, Lassig JP, Kallmes KM, Nussbaum LA. Microsurgical treatment of unruptured middle cerebral artery aneurysms: a large, contemporary experience. *J Neurosurg*. 2018;130:1498-1504.
 29. Alreshidi M, Cote DJ, Dasenbrock HH, et al. Coiling versus microsurgical clipping in the treatment of unruptured middle cerebral artery aneurysms: a meta-analysis. *Neurosurg*. 2018;83:879-889.
 30. Toccaceli G, Diana F, Cagnazzo F, et al. Microsurgical clipping compared with new and most advanced endovascular techniques in the treatment of unruptured middle cerebral artery aneurysms: a meta-analysis in the modern era. *World Neurosurg*. 2020;137:451-464.
 31. Darsaut TE, Fahed R, Macdonald RL, et al. Surgical or endovascular management of ruptured intracranial aneurysms: an agreement study. *J Neurosurg*. 2018;131:1-7.
 32. Darsaut TE, Estrade L, Jamali S, Bojanowski MW, Chagnon M, Raymond J. Uncertainty and agreement in the management of unruptured intracranial aneurysms. *J Neurosurg*. 2014;120:618-623.
 33. Schwartz D, Lellouch J. Explanatory and pragmatic attitudes in therapeutic trials. *J Clin Epidemiol*. 2009;62:499-505.
 34. Chan AW, Tetzlaff JM, Altman DG, et al. SPIRIT 2013 statement: defining standard protocol items for clinical trials. *Ann Intern Med*. 2013;158:200-207.
 35. Horton R. Surgical research or comic opera: questions, but few answers. *Lancet*. 1996;347:984-985.
 36. Raymond J, Darsaut TE, Altman DG. Pragmatic trials can be designed as optimal medical care: principles and methods of care trials. *J Clin Epidemiol*. 2014;67:1150-1156.
 37. Raymond J, Darsaut TE, Roy D. Care and research concepts should be revised to practice outcome-based medical care. *J Clin Epidemiol*. 2019;116:155-160.
 38. Benomar A, Farzin B, Volders D, et al. Angiographic results of surgical or endovascular treatment of intracranial aneurysms: a systematic review and inter-observer reliability study. *Neuroradiology*. 2021;42:1615-1620.

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